1 10. Combinatorial Designs

Combinatorial Designs

1.1 10.1. Modular Arithmetic

Section 10.1. Modular Arithmetic

Given some integer n, the 'integers modulo n' or 'integers mod n' is the set

$$Z_n = \{0, 1, \ldots, n-1\}$$

For some integer x not in Z_n, $x \mod n$ is the unique integer b in Z_n such that

$$x = cn + b$$

for some integer c.

Examples from Z_5

What is $17 \mod 5$?

Answer: Because $17 = (3)5 + 2$ we have $17 \mod 5 = 2$.

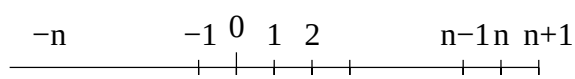
Notation

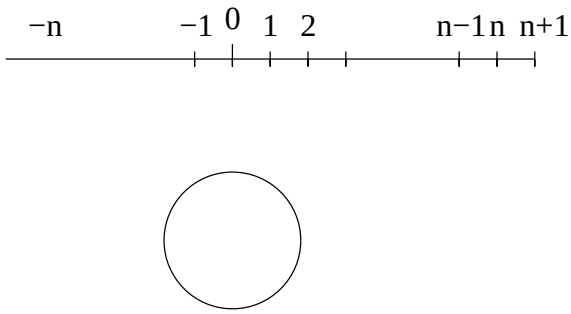
We also write $17 = 2(\mod 5)$, or just $17 = 2$ when it is clear we are in Z_5.

What is $-2 \mod 5$?

Answer: 3

A picture of Z_n

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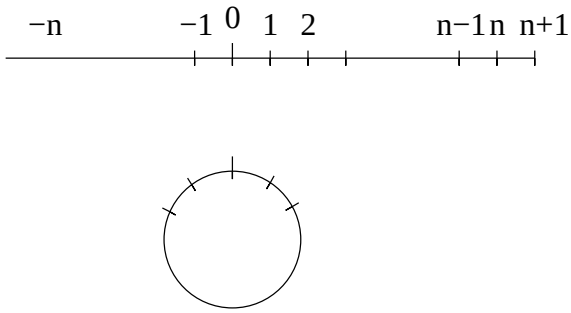
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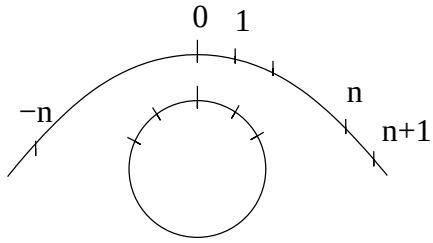
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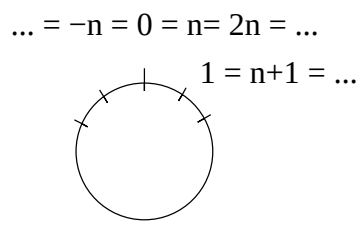


Addition, subtraction, and multiplication are inherited from Z.

Example: In Z_8,

$$3 + 7 = 10 = 2$$

$$4 \times 5 - 2 = 18 = 2$$

Notation
To differential from operations in Z, we may use $\oplus$ for $+$ and $\otimes$ for $\times$ in Z_n.

$$3 \oplus 7 = 10 = 2 \text{ in } Z_8$$

Observe that it doesn't matter if we do the operations before or after the modulo reductions:

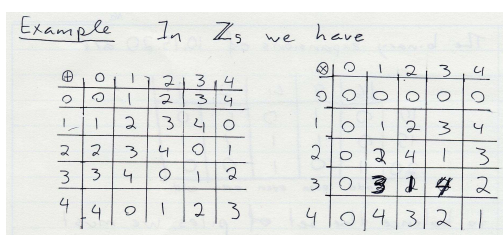
Example: In Z_{13},

$$27 \otimes 25 = 675 = 12$$

or

$$27 \otimes 25 = 1 \otimes 12 = 12$$

The operations $\oplus$ and $\otimes$ are often displayed in *addition* or *multiplication tables*.



Example In $\mathbb{Z}_5$ we have

$\oplus$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\otimes$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Inverses in Z_n

The additive inverse $-x$ of an integer x in Z_n is the integer in Z_n such that $x \oplus -x = 0$.

Example: In Z_{10}, $-7 = 3$, because

$$7 \oplus 3 = 0.$$

The additive inverse always exists and is easy to find.

Inverses in Z_n

The multiplicative inverse x^{-1} of an integer x in Z_n is the integer in Z_n such that $x \otimes x^{-1} = 1$, if it exists.

Example:

In Z_{10}

$\otimes$	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7
4	0	4	8	2	6	0	4	8	2	6
5	0	5	0	5	0	5	0	5	0	5

We see: $1^{-1} = 1, 3^{-1} = 7, 7^{-1} = 3$ and $9^{-1} = 9$, but the other integers don't have multiplicative inverses.

Multiplicative inverses and the GCD

Definition

Given two integers a, b (in Z), $\text{GCD}(a, b)$ is the greatest integer that properly divides both a and b.

Example: $\text{GCD}(15, 25) = 5$.

Theorem

An integer m has an inverse in Z_n if and only if $\text{GCD}(m, n) = 1$.

Corollary

If p is a prime integer, then every integer (except 0) has an inverse in Z_p.

The Extended GCD Algorithm

Observe that if an integer g divides positive integers a and b (for $a > b$):

$$a = gc_1 \text{ and } b = gc_2,$$

then g divides $a - b$:

$$a - b = gc_1 - gc_2 = g(c_1 - c_2),$$

It follows that if $g = \mathrm{GCD}(a, b)$ then $g = \mathrm{GCD}(a, a - b)$.

This leads to an algorithm for finding the GCD of two integers a and b, which can also be used to compute the inverse of a modulo b.

The GCD Algorithm

Algorithm
Let positive integers $a \geq b$ be given. Set $A = a$ and $B = b$. While $A > 0$ do If $A \geq B$ let $A := A - B$, else let $B := A - B$. Set $\mathrm{GCD}(a, b) = B$.

Example: Find $\mathrm{GCD}(42, 17)$. (On board).

The Extended GCD Algorithm

Backtracking through the algorithm allows us to write $\mathrm{GCD}(a, b)$ as $ca + db$ for some integers c and d.

Example: Find 17^{-1} in Z_{42}. (On board).

1.2 10.2. Block Designs

Section 10.2. Block Designs

Question
How many 3-element subsets of the set $S = \{0, 1, 2, 3, 4, 5, 6\}$ do we need to be sure that every pair of elements from S is in at least one subset?

A lower bound:

- There are $\binom{7}{2} = 21$ pairs,
- $\binom{3}{2} = 3$ in each 3-set,
- so we need at least $21/3 = 7$ 3-sets.

This is enough:

$$\{0, 1, 3\}, \{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 6\}, \{0, 4, 5\}, \{1, 5, 6\}, \{0, 2, 6\}$$

Balanced Incomplete Block Designs

More generally we will look at the question: Can we find b k-subsets of a set of v elements such that each pair occurs exactly λ times?

Let b, k and v be integers with $k \leq v$.

A block design $\mathcal{B}$ with parameters b, k and v is a a family

$$\mathcal{B} = \{B_1, \ldots, B_b\}$$

of b (not neccessarily distinct) k-subsets of a set X of v elements. The B_i are called *blocks*, The elements of X are called *varieties*,

A block design in which each pair of elements of X occurs in exactly λ blocks is called a *balanced* block design of index λ.

Block designs are often represented compactly by their incidence matrix . This is an $b \times v$ matrix whose rows are indexed by the blocks, and whose columns are indexed by the varieties. The i, j^{th} entry is 1 if the jth variety is in the ith block, and is 0 otherwise.

Example: The design could be represented by

$$\{0,1,3\},\{1,2,4\},\{2,3,5\},\{3,4,6\},\{0,4,5\},\{1,5,6\},\{0,2,6\}$$

	0	1	2	3	4	5	6
B_1	1	1	0	1	0	0	0
B_2	0	1	1	0	1	0	0
B_3	0	0	1	1	0	1	0
B_4	0	0	0	1	1	0	1
B_5	1	0	0	0	1	1	0
B_6	0	1	0	0	0	1	1
B_7	1	0	1	0	0	0	1

A trivial block design with parameters $v = k = 5$ and $b = 3$ is

$$\begin{aligned} B_1 &= \{1,2,3,4,5\} \\ B_2 &= \{1,2,3,4,5\} \\ B_3 &= \{1,2,3,4,5\} \end{aligned}$$

A block design is complete if $v = k$. These are uninteresting: each block is just X. We consider incomplete designs, when $k < v$.

In the above example, each variety occurs in 3 blocks. This is called the repetition of the varieties, and is denoted by r.

Definition
A BIBD with parameters b, v, k, r, λ is a Balanced Incomplete Block Design with v varieties, b blocks of size k, and index λ, in which each variety occurs in r blocks,

For what values of the parameters b, v, k, r, λ, do BIBDs exist?
Necessary Conditions

- $2 \leq k < v$ (By definition.)
- $r(k-1) = \lambda(v-1)$ Thus $\frac{\lambda(v-1)}{k-1}$ must be an integer.
- $bk = vr$

- $\lambda < r$
- $v \leq b$ (Fisher's inequality)

(Prove 2nd and 3rd on board.)

Proof of Fisher's inequality: $v \leq b$**.**

$$\begin{array}{c} v_i \\ \boxed{1} \end{array}$$

Start with the $b \times v$ incidence matrix A. Consider the matrix $A^T A$.

Now

$$\left|\begin{bmatrix} r & \lambda & \lambda & \lambda \\ \lambda & r & \lambda & \lambda \\ \lambda & \lambda & r & \lambda \\ \lambda & \lambda & \lambda & r \end{bmatrix}\right| = \left|\begin{bmatrix} r & \lambda & \lambda & \lambda \\ \lambda-r & r-\lambda & 0 & 0 \\ \lambda-r & 0 & r-\lambda & 0 \\ \lambda-r & 0 & 0 & r-\lambda \end{bmatrix}\right| = \left|\begin{bmatrix} r+3\lambda & \lambda & \lambda & \lambda \\ 0 & r-\lambda & 0 & 0 \\ 0 & 0 & r-\lambda & 0 \\ 0 & 0 & 0 & r-\lambda \end{bmatrix}\right|$$

So $A^T A$ is non-singular.
As it is $v \times v$ it has rank v.
So A has rank at least v.
But A is $b \times v$, so $v \geq b$.

Definition
A BIBD with $b = v$ is symmetric , and called a SBIBD for short.

When $b = v$, we also have $k = r$ and λ is determined by

$$\lambda = \frac{k(k-1)}{v-1}.$$

For what values of b and k are there $SBIBD$s: block designs with

- b blocks, and b varieties,
- blocks of size k and repetition k,
- each pair in $\lambda = \frac{k(k-1)}{b-1}$ blocks?

An SBIBD with $b = 7$, $k = 3$

In this case we have $\lambda = 3(2)/6 = 1$ so each pair of varieties occurs only once. Let the varieties be $Z_7 = \{0, 1, 2, 3, 4, 5, 6\}$.. We find 7 blocks of size 3 such that each pair of integers occurs exactly once.

- Start with this block.
- Build the other blocks by adding the same number (modulo 7) to each entry:
- Check that each pair occurs exactly once.

$$\begin{aligned} B_1 &= \{0, 1, 3\} \\ B_2 &= \{1, 2, 4\} \\ B_3 &= \{2, 3, 5\} \\ B_4 &= \{3, 4, 6\} \\ B_5 &= \{4, 5, 0\} \\ B_6 &= \{5, 6, 1\} \\ B_7 &= \{6, 0, 2\} \end{aligned}$$

Why this works, (and checking the pairs)

$$\begin{aligned} B_1 &= \{0, 1, 3\} \\ B_2 &= \{1, 2, 4\} \\ B_3 &= \{2, 3, 5\} \\ B_4 &= \{3, 4, 6\} \\ B_5 &= \{4, 5, 0\} \\ B_6 &= \{5, 6, 1\} \\ B_7 &= \{6, 7, 2\} \end{aligned}$$

In every block,

- the first variety occurs with varieties 1 and 3 greater than it,
- the second variety occurs with varieties 2 and 6 greater than it, and
- the third variety occurs with varieties 4 and 5 greater than it.

Since each variety occurs once in each position, each variety occurs exactly once with each other variety.

Let B be a k-set of Z_v such that the following is true.

All non-zero elements of Z_v occur as the difference between exactly λ pairs of elements of B.

(Such a set B is called a difference set .)

Then there exists a SBIBD $\{B_1, \ldots, B_v\}$ with parameters k, v and $\lambda = \frac{k(k-1)}{l-1}$ constructed by letting, for $i = 1, \ldots, v$, $B_i = i + B$ the block in we add i to each eelment of B.

1.3 10.3. Steiner Triple Systems

Section 10.3. Steiner Triple Systems.

A BIBD with $k = 3$ is called a Steiner Triple System .

We look at those with index $\lambda = 1$.

(If we get a system with index 1 we can get a system with index λ by taking λ copies of each block.)

So we look for families of 3-sets of a Z_v, such that every pair of elements is in exactly one 3-set.

Identities

From $r = \frac{\lambda(v-1)}{k-1}$ we have

$$r = \frac{(v-1)}{2}$$

and from $bk = vr$ we have

$$b = \frac{v(v-1)}{6}.$$

Thus v is odd, and $v(v-1) = 0 \mod 6$ (ie. $v(v-1)$ is divisible by 6).

In Z_6,

v	$v(v-1)$
1	0
3	0
5	2

So $v = 6n + 1$ or $6n + 3$ for some integer n.

Theorem

Steiner Triple Systems exist for $v = 3$, and $v = 6n + 1$ and $v = 6n + 3$ for any positive integer n.

So they exist for v in

$$3, 7, 9, 13, 15, 19, 21, 25, 27, 31, 33, 37, 39...$$

We don't prove this, but we show how to get an infinite number of them.

A Steiner Triple System with $v = 3$:

$$\{0, 1, 2\}.$$

A Steiner Triple System with $v = 7$:

$$\{0, 1, 3\}, \{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 6\}, \{0, 4, 5\}, \{1, 5, 6\}, \{0, 2, 6\}$$

A Construction of Steiner Triple Systems

Given two Steiner Triple Systems, with v and w varieties respectively, we construct a new Steiner Triple System with vw varietes.

This gives us Steiner Triple Systems with varieties of size

$$3, 7, 9, 13, 15, 19, 21, 25, 27, 31, 33, 37, 39...$$

(Construction on board.)

1.4 10.4. Latin Squares

Section 10.4. Latin Squares

Recall that a latin square of order n is an $n \times n$ matrix A with entries in Z_n such that each integer mod n occurs exactly once in each row and each column:

Example:

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 4 & 0 & 1 & 2 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 2 & 3 & 4 & 0 & 1 \\ 1 & 2 & 3 & 4 & 0 \end{bmatrix}$$

We index the rows and columns with Z_n, so have 0th to $n-1^{th}$ rows. Thus $A_{1,2}$ is 1.

Given a latin square of order n, we can always get other ones by permuting the entries:

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 4 & 0 & 1 & 2 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 2 & 3 & 4 & 0 & 1 \\ 1 & 2 & 3 & 4 & 0 \end{bmatrix}$$

Switching $0 \to 3$, $1 \to 1$, $2 \to 0$, $3 \to 2$ and $4 \to 4$, we get

So we may always assume that the 0th row is $0, 1, 2, \ldots, n-1$, by permuting entries. We call this the standard form .

The simple example

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 & 0 \\ 2 & 3 & 4 & 0 & 1 \\ 3 & 4 & 0 & 1 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{bmatrix}$$

obviously generalises to a latin square of order n for every n. It is the addition table of Z_n.

This is defined succinctly by $A_{i,j} = i + j \mod n$.

We generalise this construction to get different latin squares of order n. First we need a lemma.

Lemma
If r has a multiplicative inverse in Z_n, then the n integers $r, 2r, 3r, \ldots, nr$ are distinct in Z_n.

Proof?

If $ar = br$ then $a = arr^{-1} = brr^{-1} = b$.
For any non-zero $r \in Z_n$, let L_n^r be the matrix defined by

$$(L_n^r)_{i,j} = r \times i + j \mod n.$$

$$\text{Example } L_5^2\text{:} \qquad \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 0 & 1 \\ 4 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 & 0 \\ 3 & 4 & 0 & 1 & 2 \end{bmatrix}$$

- Every row has n distinct elements.
- If r has a multiplicative inverse, then every column has n distinct elements.

Theorem
Let n be a positive integer, and $r \in Z_n$ be such that $GCD(n, r) = 1$. Then L_n^r is a latin square.

Recall that two order n latin squares A and B are orthogonal if the pairs

$$(A_{i,j}, B_{i,j}),$$

for all $i, j \in Z_n$, are distinct.

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 4 & 0 & 1 & 2 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 2 & 3 & 4 & 0 & 1 \\ 1 & 2 & 3 & 4 & 0 \end{bmatrix} . \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 0 & 1 \\ 4 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 & 0 \\ 3 & 4 & 0 & 1 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} (0,0) & (1,1) & (2,2) & (3,3) & (4,4) \\ (4,2) & (0,3) & (1,4) & (2,0) & (3,1) \\ (3,4) & (4,0) & (0,1) & (1,2) & (2,3) \\ (2,1) & (3,2) & (4,3) & (0,4) & (1,0) \\ (1,3) & (2,4) & (3,0) & (4,1) & (0,2) \end{bmatrix}$$

Some History

- Euler (about 1760) conjectured that there are pairs of Orthogonal Latin Squares of order n for $n \neq 2, 6, 10, 14, \ldots$.
- *They were known to exist for all other values.*
- Tarry (1901) proved there are none for $n = 6$.
- Bose, Parker, Shrikhande, (1960) showed they exist for all $n > 6$.

We see the existence of such pairs for $n \neq 2, 6, 10, 14, \ldots$.

Orthogonal Latin Squares of Odd Order n

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 4 & 5 & 6 & 0 \\ 2 & 3 & 4 & 5 & 6 & 0 & 1 \\ 3 & 4 & 5 & 6 & 0 & 1 & 2 \\ 4 & 5 & 6 & 0 & 1 & 2 & 3 \\ 5 & 6 & 0 & 1 & 2 & 3 & 4 \\ 6 & 0 & 1 & 2 & 3 & 4 & 5 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 5 & 6 & 0 & 1 \\ 4 & 5 & 6 & 0 & 1 & 2 & 3 \\ 6 & 0 & 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 & 6 & 0 \\ 3 & 4 & 5 & 6 & 0 & 1 & 2 \\ 5 & 6 & 0 & 1 & 2 & 3 & 4 \end{bmatrix}$$

Why are L_n^1 and L_n^2 orthogonal for every odd n ?

For prime $n \neq 2$ we can do better.

A set of k latin squares $A_1, \ldots, A_k$ such that any two are orthogonal is called a set of Mutually Othogonal Latin Squares , or MOLS .

Theorem

If $A_1, \ldots A_k$ is a set of MOLS of order n, then $k \leq n - 1$.

Theorem

If n is prime, then $L_n^1, L_n^2, \ldots, L_n^{n-1}$ are $n - 1$ MOLS.

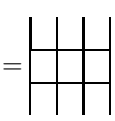
(Proofs.)

One can also show that if $n = p^d$ for $d \geq 1$ then there are $n - 1$ MOLs of order n. (This even works for $p = 2$.) We won't do this.

A multiplicative construction

For Latin Squares A and B of orders a and b respectively, we can construct a Latin Square $A \otimes B$ of order ab as in the following example.

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \oplus \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{bmatrix}$$

$=$ 

It isn't hard to check that if A and A' are orthogonal, and B and B' are orthogonal, then $A \otimes A'$ and $B \otimes B'$ are orthogonal.

So using that there are pairs of orthogonal Latin Squares of order n for

- n being odd, and
- $n = 2^d$ for $d \geq 2$ (ie, $4, 8, 16, 32, \ldots,$)

We get that there are pairs of orthogonal Latin Squares of order n for

$$\begin{aligned} n &\neq 2, 2(3), 2(5), 2(7), \ldots, \\ &= 2, 6, 10, 14, \ldots, \end{aligned}$$

This is what Euler knew.